

Module 5

Information Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

A direct cutover implementation strategy means that the old system is turned off and then the new system is turned on at a particular point in time.

However, a side-by-side implementation strategy means that both the old and new systems operate simultaneously for a time until confidence in the new system is established.

Answer 1(b)

A side-by-side implementation strategy is the least risky because if the new system is found to have serious flaws, the project manager may consider switching back all the operations to the old system until those flaws are addressed and solved.

Answer 1(c)(i)

Accountants' duties in the testing stage include the followings:

- participating in alpha testing and beta testing;
- ensuring real data from all sources is entering into the system correctly;
- ensuring the system is processing real data correctly to produce accurate outputs; and
- catching any errors and reporting them to programmers.

[Any two of the above]

Answer 1(c)(ii)

Accountants' duties in the operation and maintenance stage include the followings:

- monitoring the new system on an ongoing basis;
- determining whether updates are necessary to adapt to changes of business needs, regulations and/or accounting standards; and
- alerting the programming team about the bugs or system errors of the new system.

[Any two of the above]

Answer 1(d)

An organisation can support a system change through revising management policies in the following ways

- updating the formal standard operating procedures, such as requiring all staff to use the new system when carrying out routine daily operations;
- revising staff performance measurements and rewards to motivate all staff to use the new system, such as by giving early adopters a higher multiplier (for example, an additional 20%) of the total score of a specific performance evaluation area, or by giving them a higher opportunity to receive a cash bonus or a share option as an award; and
- reallocating more resources to communicate the importance of adopting the new system, such as redirecting funds and staff to the project team which can provide additional supports and trainings to users.

[Any two of the above]

Answer 2(a)(i)

Biometric authentication means that a person's physical characteristic is used for his/her identity verification.

Answer 2(a)(ii)

The following biometric factors can be demonstrated by a fingerprint authentication system:

- universality: humans have fingers and each finger has a fingerprint ;
- uniqueness: fingerprints are unique, and no one has the exact same set of ridges and lines that you have on your fingers;
- permanence: a person's fingerprints remain essentially unchanged throughout his/ her lifetime;
- collectability: it is easy to acquire fingerprints from users for authentication;
- performance: fingerprint scanners and their identification systems work accurately and quickly nowadays; and
- acceptability: people are willing to use fingerprints in some situations, such as for room access, employee attendance, or immigration clearance.

[Any three of the above]

Answer 2(b)(i)

A hacker can set up a wireless router or a wireless access point that looks legitimate to an organisation in a location with weak WiFi signal coverage. If a user connects his/her device to this wireless router or access point, all the communication across the router or access point can be intercepted by the hacker.

Answer 2(b)(ii)

A hacker can send some phishing e-mails with a malicious hyperlink or an attachment of a specific malware which targets common web browsers. If a user clicks on the malicious hyperlink or executes the attachment directly without any consideration, the web browser can be compromised and used to intercept or even modify data/information that would be transmitted to a web server.

Answer 2(c)(i)

A virtual private network (VPN) is a secure network communication that is established between the computers and servers across a shared network or the Internet.

Answer 2(c)(ii)

Once a VPN is established, all the data transmissions will be encrypted and decrypted automatically at a user's device and a secure external computer/ server respectively or vice versa. Even if a user connects his/her device to a wireless router or access point controlled by a hacker and uses it for data transmission, the hacker cannot understand what is being transmitted.

Answer 3(a)

- (i) Strategic level
- (ii) Tactical level
- (iii) Operational level
- (iv) Operational level
- (v) Strategic level
- (vi) Tactical level

Answer 3(b)

- (i) A transaction processing system processes data from both financial and non-financial business transactions.
- (ii) A management information system extracts and summarises data from transaction processing systems to support routine management activities.
- (iii) An expert system mimics a human expert in a particular area and makes decisions.
- (iv) An executive information system presents structured, summarised information about aspects of the business operations important to executives.

Answer 3(c)

Organisations may face the following types of ethical and legal issues in using information systems:

- privacy issues – organisations collect and use data about individuals must comply with relevant privacy laws and regulations;
- data accuracy issues – wrong data may cause a mistake in decision making or providing services which may lead to legal liability;
- property issues – organisations collect a large volume of personal information, but they have uncertainties about who can own such personal information or even earn profit from using such personal information; and
- accessibility issues – organisations need to decide who can access their information in different situations and conditions.

[Any three of the above]

Answer 4(a)(i)

Information needs of human resources include the following:

- applicant records, such as application forms and resumes;
- list of positions among the departments within the organisation;
- list of skills needed per each job position;
- fringe benefits information; and
- relevant labour laws and policies.

[Any two of the above]

Answer 4(a)(ii)

Information needs of marketing include the following:

- market research or competitor research per each type of product and services;
- purchase cost or competitor price per each product;
- customers' preferences and their consumption behaviours; and
- media or website statistics for promotion.

[Any two of the above]

Answer 4(b)(i)

The differences between a server and a client are as follows:

- a server is a network-connected physical or virtual device dedicated to providing services or resources to the rest of the network, while a client is a device that accesses a server;
- a server usually has more processing power, memory, and storage capacity than a client;
- a server is usually used for processing complex computations and the central storage of data, but a client is usually used for data input and presenting system outputs; and
- a server supports multiple user connections and allows users to use its resources simultaneously, but a client usually supports a single user login only.

[Any three of the above]

Answer 4(b)(ii)

A network switch acts as a controller and enables connected devices to communicate with each other.

It is used to build a network and connect computers, servers, printers, wireless access points, and/or other devices together within an office or a building.

Answer 5(a)

The following risks may occur if an e-commerce business has weak identity authentication among its customers and suppliers:

- providing products, services, or payments to individuals who are not whom they claim to be;

- becoming an unwitting accomplice in criminal activities, such as money laundering;
- providing system access to hackers who may damage the system or steal sensitive information, such as customer information (including payment details) and company secrets; or
- entering into contractual agreements with parties who later claim they never signed the contract.

[Any two of the above]

Answer 5(b)

The differences between e-commerce and e-business are as follows:

- e-business is broader than e-commerce. Other than buying and selling goods and services, e-business includes electronically servicing customers, collaborating with business partners, and performing internal transaction-based processes such as supply chain management (SCM) and human resources management (HRM);
- e-business involves the organisation's internal back-end operations, but e-commerce applies exclusively to front-end operations;
- e-business includes e-commerce, but it has the additional goal of improving efficiency and productivity, reducing costs within an organisation; and
- organisations typically conduct e-business with intranets and extranets, but they conduct e-commerce through the Internet.

Answer 5(c)(i)

Robotic Process Automation (RPA) is the application of a software robot for automation of manual tasks that do not require human knowledge.

Answer 5(c)(ii)

Examples of using RPA are as follows:

- cross-checking purchase invoices with purchase orders and goods received notes;
- inputting payment vouchers based on the information provided by the supporting documents;
- routing transactions for approval;

- preparing a bank reconciliation by matching transactions between a bank statement and a bank ledger account;
- automating pre- and/or post- payment validation; or
- monitoring and identifying potential duplicate payments,

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